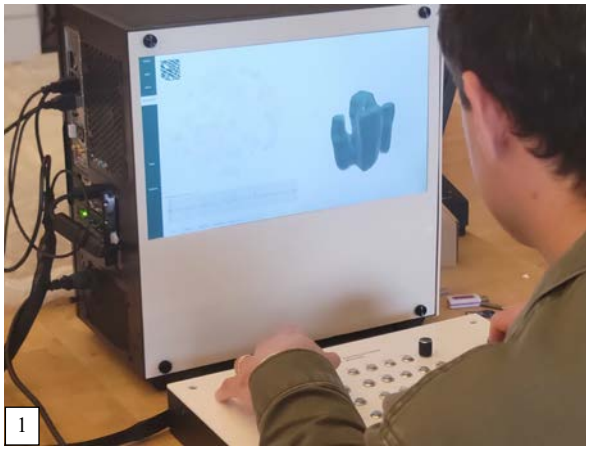
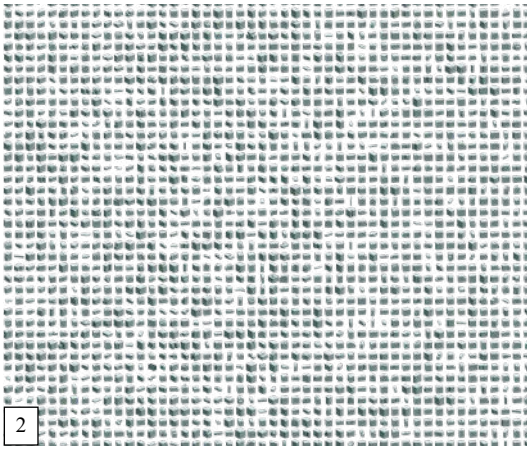
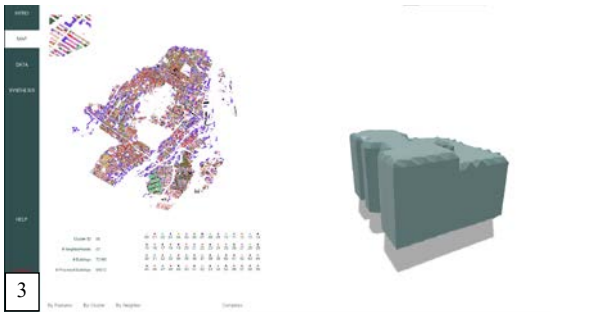
BUILDING FORM ANALYZER

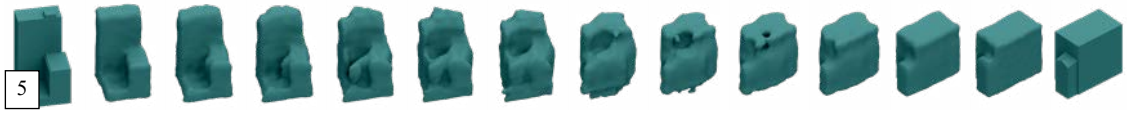
An Inductive Method for Classifying and Generating Building Form in a City with Implications for Orientation

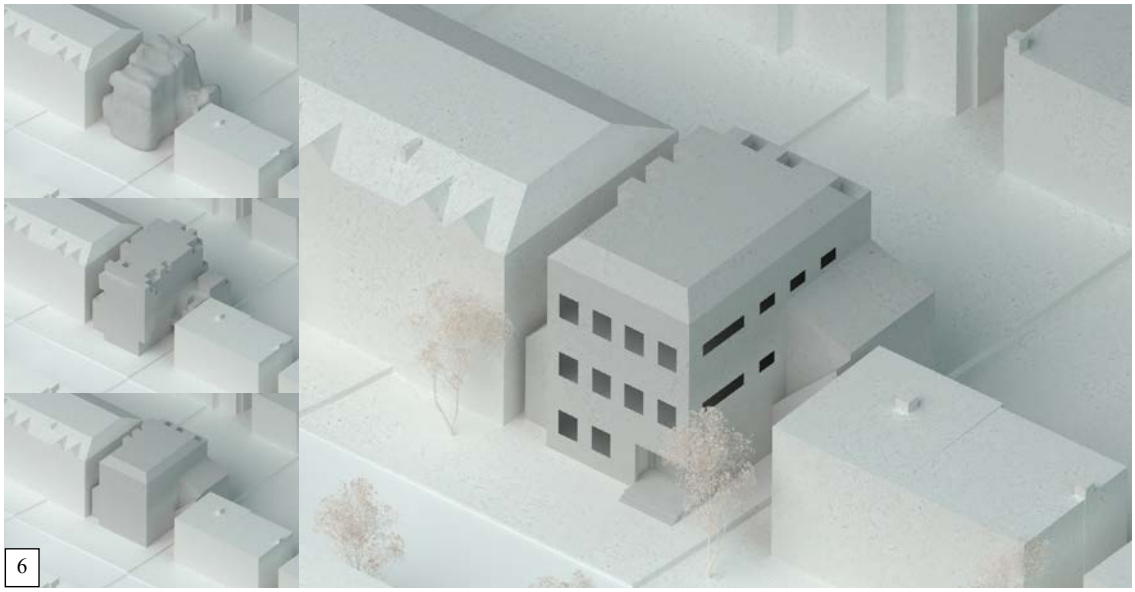
Video	https://vimeo.com/898223212
Details	https://jinmorhee.net/jinmorhee_3-multimedia/montrealbuildingtypes.html
Published	Journal of Envrionmental and Planning B
Funded	Architectural Design Human Resource Development Project, Korea Agency for Infrastructure Technology Advancement
Keywords	Typo-morphology, Form Study, Deep Learning, Form Synthesis











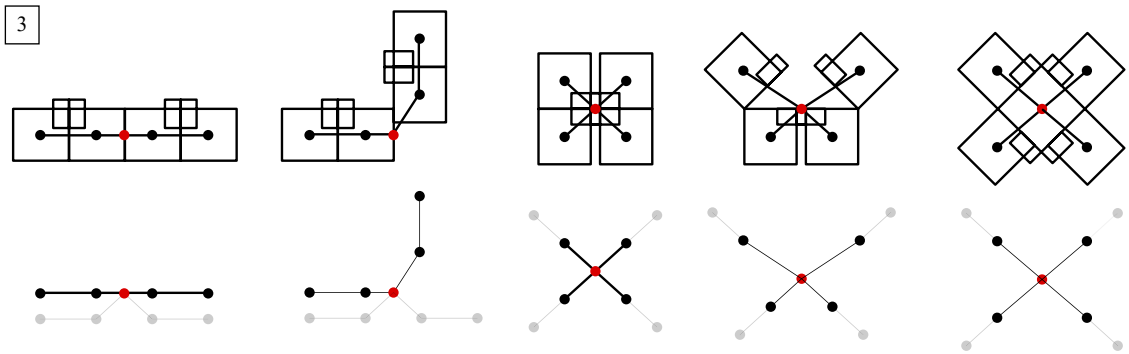
This research presents an exploration with a data-scientific method for expanding morphological understanding of city space from a large dataset of building forms. This is a computational analysis based on a pipeline using a deep neural network adapting from other morphological studies, the processing of large form datasets. I selected a dataset of building forms in Montreal for a case study. By this pipeline, I was able to identify typological traits of buildings according to their morphological features and geospatial distribution to yield new descriptors of building form types. These types are then used to analyze the morphological characteristics of an urban space. In this way, I can demonstrate the potential of the pipeline for urban morphology.

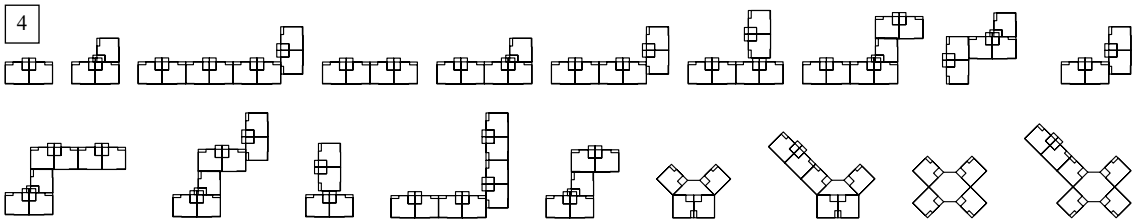
1. Dedicated Machine for Building Analyzer Montreal
2. 3D Building Dataset of Montreal
3. Screen Capture of Building Types Analysis Mode and
4. Synthesis Mode in Building Analyzer Montreal Interface
5. Series of interpolated forms between two buildings
6. Massing Design based on a Montreal Vernacular from Deep Learning Model

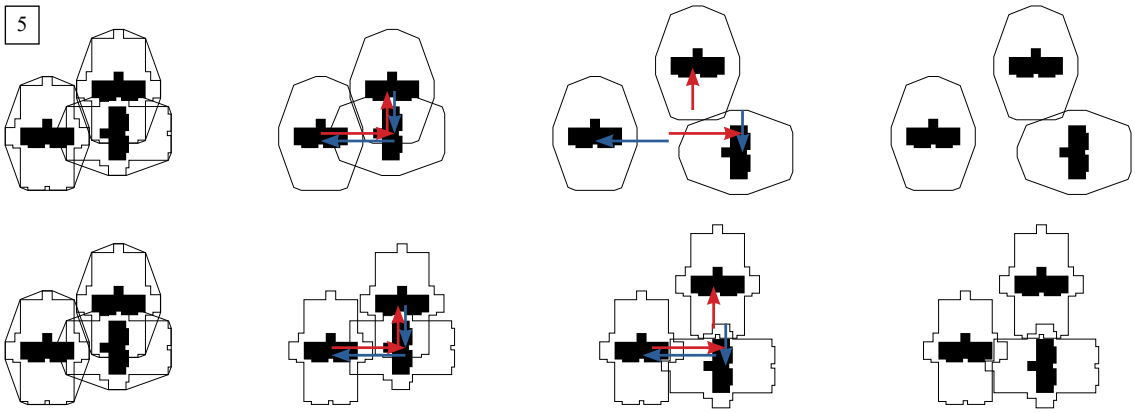
ABLi

Apartment Building Layout intelligence

Video	https://drive.google.com/file/d/1kzq-3M-eSKloDfe7ohZZj-ZXbSYryZkZ/view?usp=drive_link
Published	The 20th International Conference of CAAD Futures 2023, *INTERCONNECTIONS: Co-computing beyond boundaries*
Funded	DL E&C
Keywords	Generative Design, Physics Simulator, Optimization, Layout, Apartment Buildings







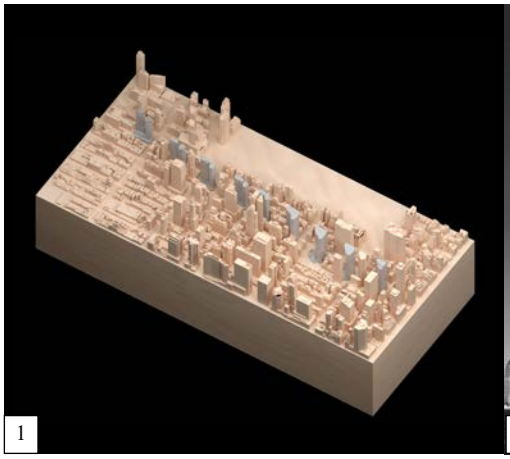
Apartment Building Layout Intelligence (ABLi) is a software to automate the layout design of an apartment complex. The software runs either as a stand-alone program or as a plug-in of Grasshopper in Rhinoceros 3D. This automation tool proposes a computational method to address the generation of realistic apartment complex layouts with heterogeneous and non-convex buildings with variable heights and setbacks in custom sites. We define this layout optimization problem as finding the optimal locations and angles of the building with design constraints. To avoid exploring the large design solution space, we introduce a series of mechanisms, such as the discretization of possible building locations and a constraint solver based on rigid-body simulation.

1. Screen Capture, Stand-alone Version of ABLi
2. Generated Apartment Buildings and and Automated Layout Example
3. Graph-based Topological Representation of Apartment Buildings
4. Various Building Shapes from the Graph-based Representation
5. Two Different Level of Physics Action-Reaction: Convex Hulls and Offset Bounds

DEEPRISE

Typological Analysis and Generation of High-rise Building Using Deep Learning

Video	https://vimeo.com/409341878
Details	https://jinmorhee.net/jinmorhee_3-multimedia/deepriseparti.html
Published	The 19th International Conference of CAAD Futures 2021, *Design Imperatives: The Future is Now*
Award	Grand Prize in Digital Architecture Competition 2021, Korea Ministry of Land, Infrastructure and Transport
Keywords	Deep Learning, Data-Driven Design, Architectural Typology, High-Rise, Form Synthesis, Computeraided Architectural Design



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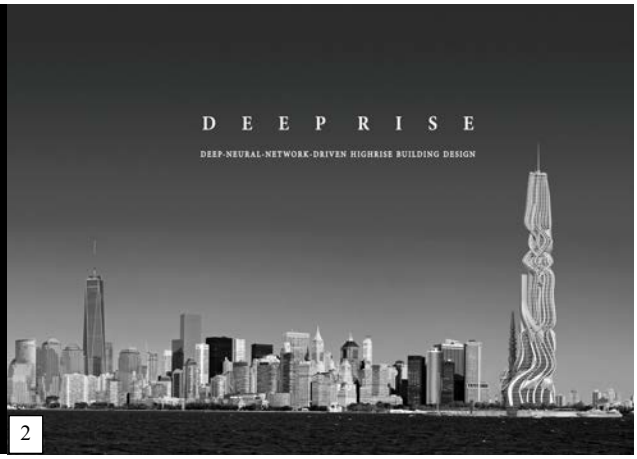

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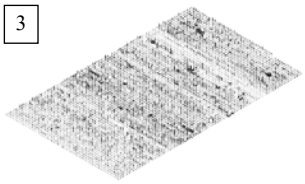

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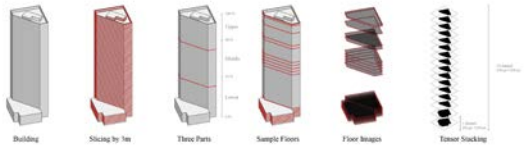

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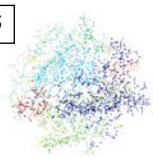

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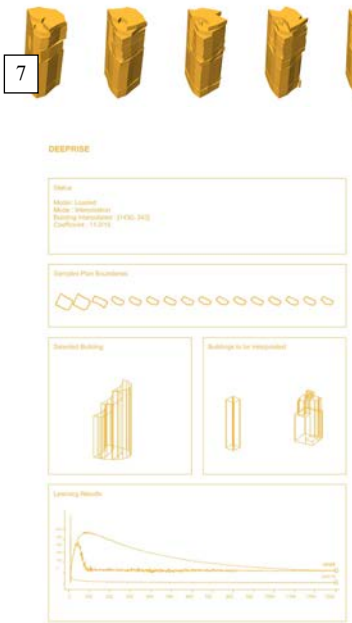

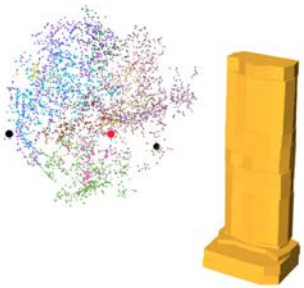

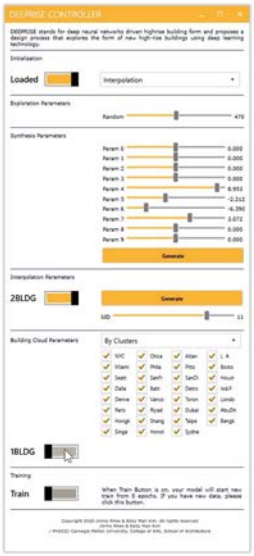

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This research proposes a new way to model and design a high-rise building form by learning the morphological features that correspond to its own architectural design principles through deep neural networks. By collecting and training diverse forms of high-rise building data from major cities in the world, we train a deep generative model and prototype a system for generating a high-rise building form in three different ways: exploration, synthesis, and interpolation. This research demonstrates that a deep generative model can grasp highly complex principles of architectural form and that design decisions for new forms can be processed based on deep learning of data. By further developing a generated form into schematic design, the system provides an example of a high-rise building design with a style distinctive to the deep generative model.

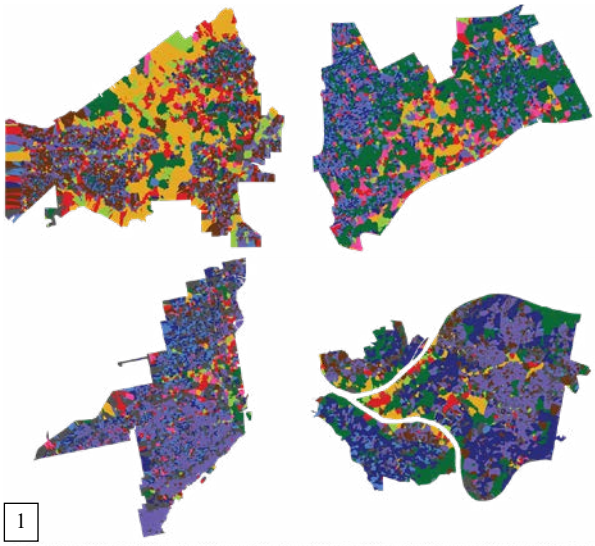
1. Design Example, Interpolation Concept
2. In-between form of Empire State Building and One World Trade Center
3. 4,596 3D Model of High-rise Buildings from 31 Cities
4. Process of Converting Building into a Tensor
5. Latent Space by Typical Forms of High-Rise Building Types
6. Series of interpolated forms between two buildings
7. Series of synthesized forms by slight changes of Z-vectors
8. A Screenshot of Deeprise Prototype

AI-DRIVEN URBAN STRUCTURE ANALYSIS

Computational Methods for the Identification and Comparative Analysis of Urban Forms: A Case Study of Rust Belt Cities

Published Doctoral Dissertation

Keywords Urban Form, Deep Learning, Classification, Urban Data



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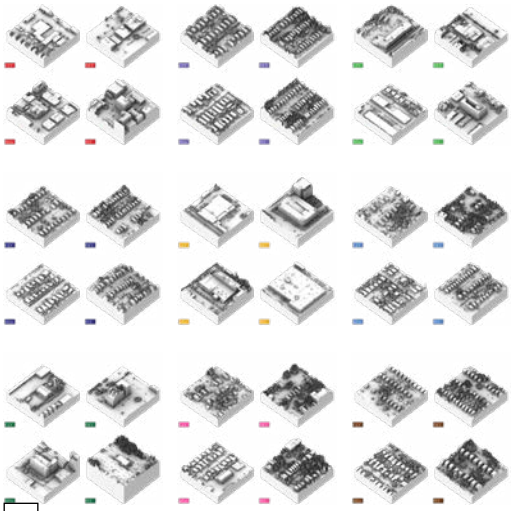

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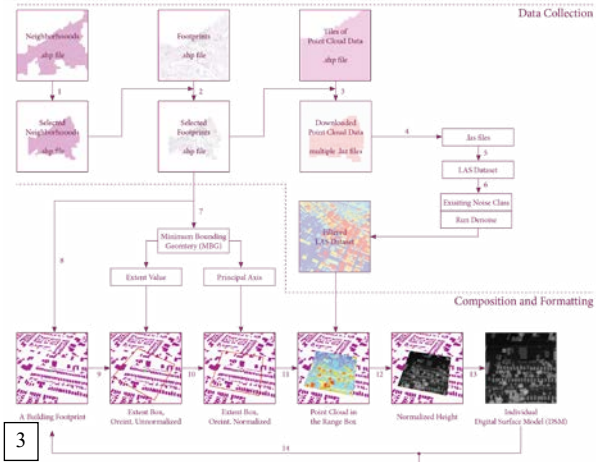

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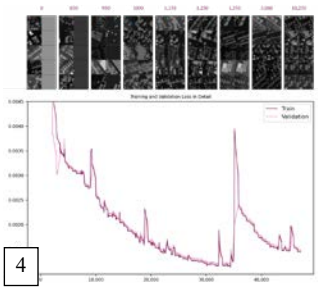

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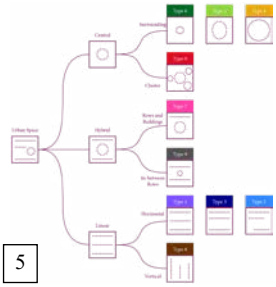

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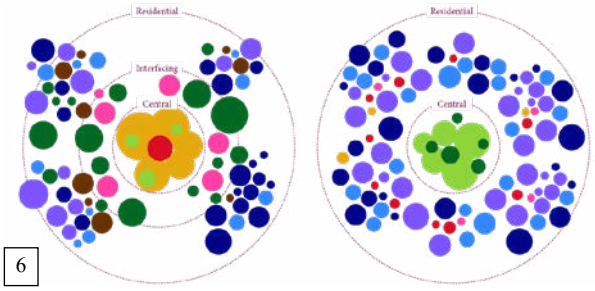

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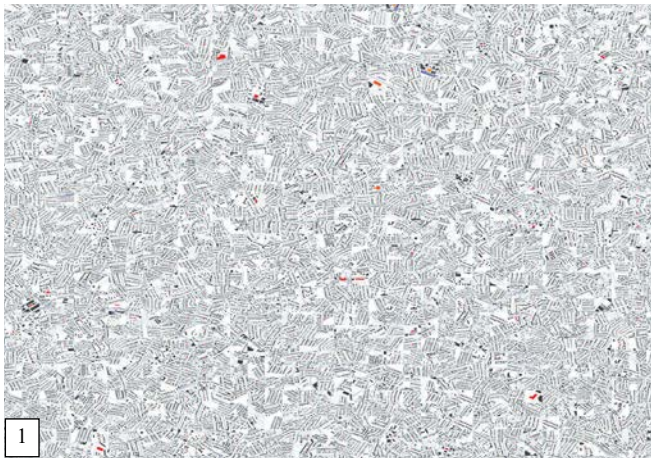
1. Maps of Classified Urban Types from a New Urban Form Dataset of Cleveland, Detroit, Pittsburgh, and Miami
2. Representative Forms
3. An Overview of Formatting and ConstruDataset
4. Graphs Depicting Training and Validation Loss
5. Inferred Urban Taxonomy based on the Identified Urban Form Types
6. A common urban structure shared among the three Rust Belt cities (left), distinguished from such structure in Miami (right)
7. Data Validation by Segmentation of Site Survey Photos

CONTEXT-RICH URBAN ANALYSIS

A Case Study in Pittsburgh, PA

Video	https://vimeo.com/348976321
Details	https://jinmorhee.net/jinmorhee_3-multimedia/contextrichurban.html
Published	The 37th eCAADe and 23rd SIGraDi Conference, *Architecture in the Age of the 4th Industrial Revolution*
Exhibited	Video work, Austria Pavilion, Vennice Biennale, 2021
	The 2nd Special Exhibition for Korean Student in America by Korean Institute of Architects, Seoul, Korea, 2019
Keywords	Urban Morphology, Machine Learning, Architectural Contexts, Urban Analysis, GIS



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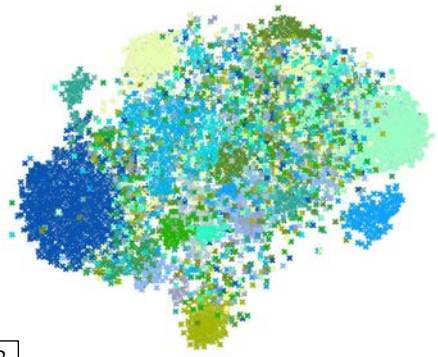

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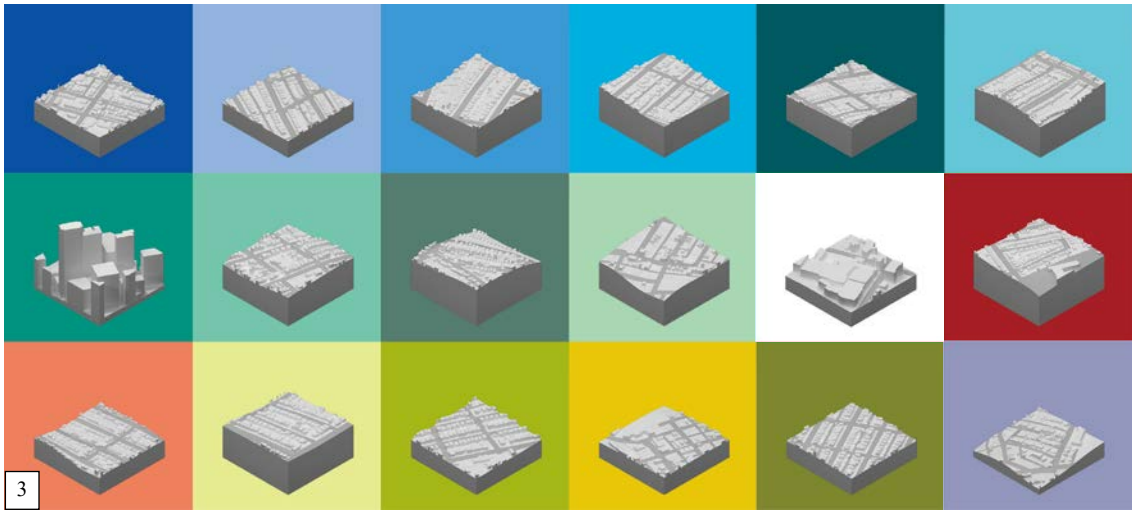

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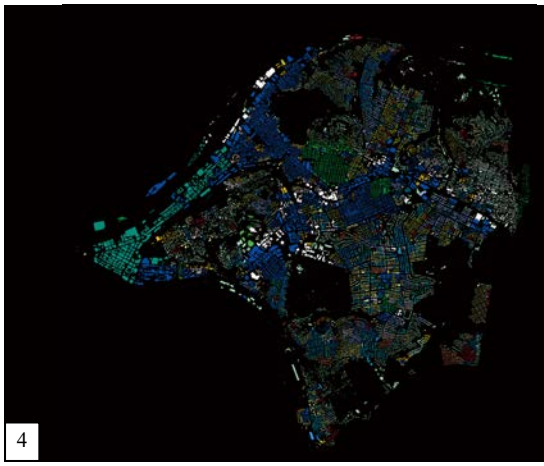

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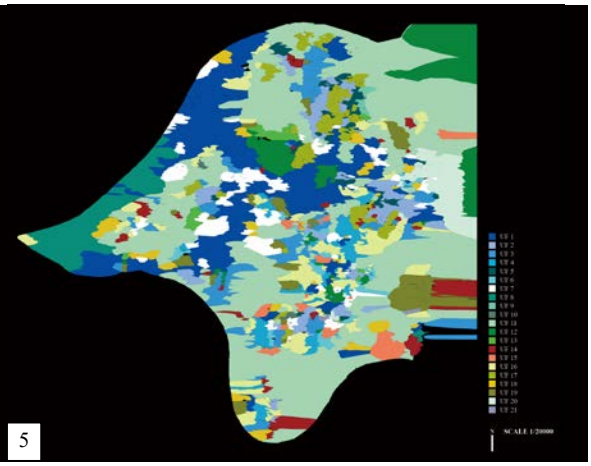

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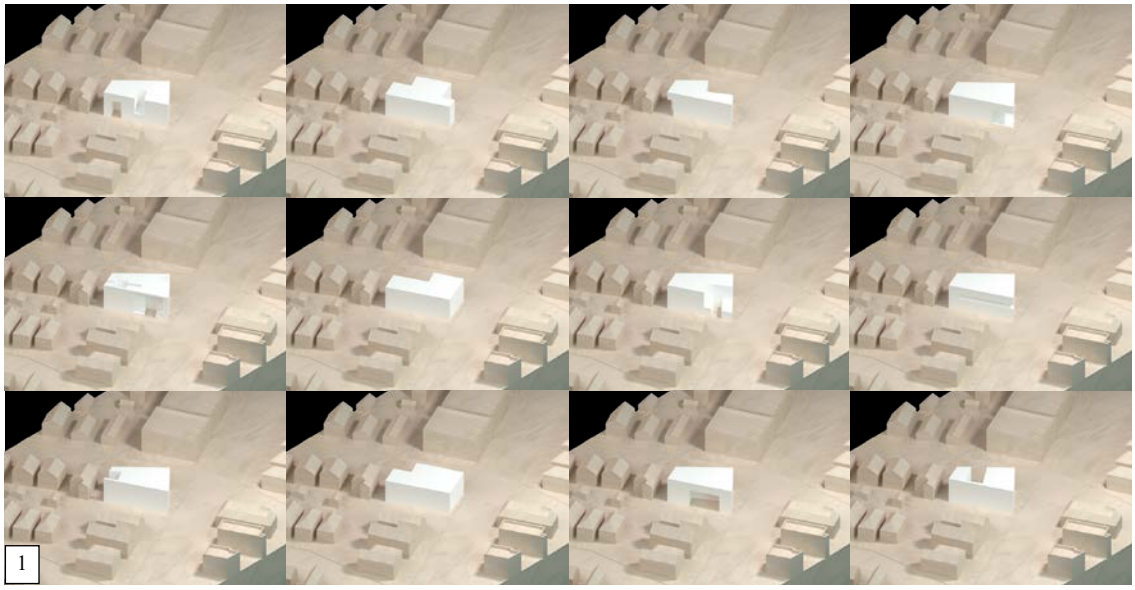
This research reports on the analytic and generative potential of using machine learning methods on context-rich urban datasets. Working with a dataset of Pittsburgh's urban data, this research proposes a novel method to extract information about each building in the city in ways that reflect its immediate urban context and analyze it in terms of urban homogeneity through machine learning algorithms. The method can identify clusters of urban spaces sharing similar architectural context conditions by statistically calculating features of all building footprints and urban contexts in Pittsburgh. From the clustering results, a new type of map called a 'data-driven map of urban morphological types' can be generated. The interpretations and applications of this map and method will be discussed. From these discussions I will address values, meanings, and potentials of this new method in terms of architectural design, urbanism, and computation.

1. Data Samples of Diagrammatic Image Dataset for PGH
2. Clustering Results of Buildings in Data Space
3. Typical Form of Each Urban Type
4. Data-driven Map of Morphological Types by Building Footprints
5. Data-driven Map of Morphological Types by Boundaries

BUILDING FOOTPRINT PREDICTION

Integrating Building Footprint Prediction and Building Massing

Details	https://jinmorhee.net/jinmorhee_2-multiimage/footprint.html
Published	The 25th International Conference of CAADRIA, *RE: Anthropocene*
Exhibited	Video work, Austria Pavilion, Vennice Biennale, 2021
	The 2nd Special Exhibition for Korean Student in America by Korean Institute of Architects, Seoul, Korea, 2019
Keywords	Urban Morphology, Machine Learning, Architectural Contexts, Urban Analysis, GIS



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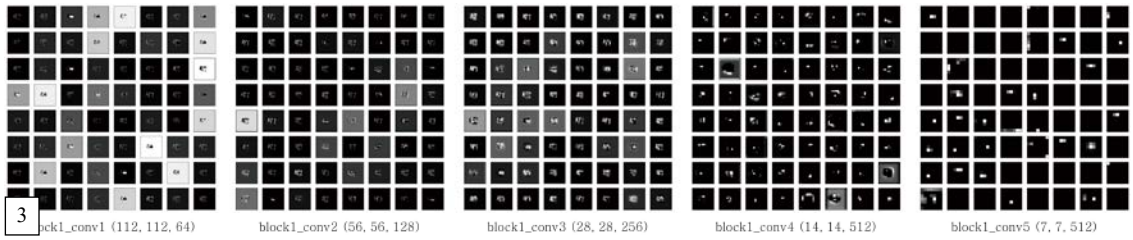

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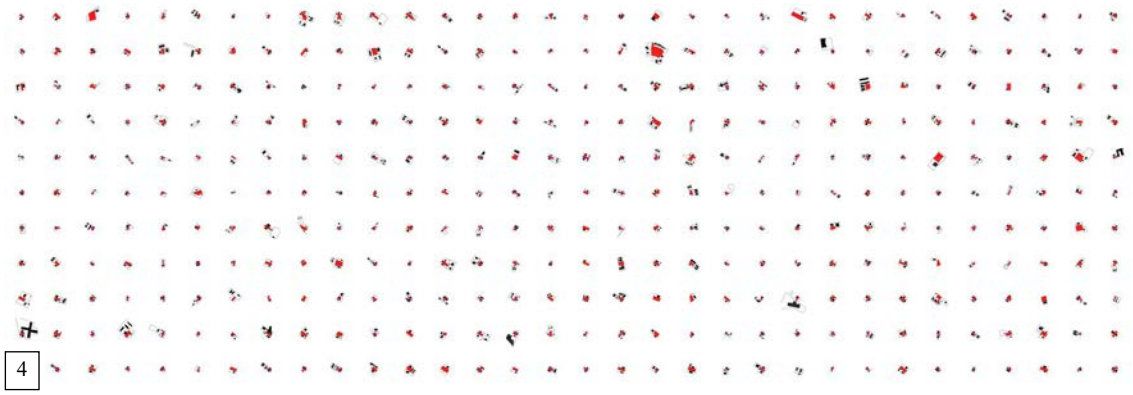

4

This research introduces a deep learning approach for generating building geometry based on contextual geometry in urban settings, focusing on building footprints as a key interface between urban and architectural forms. Using GIS data from Pittsburgh, a VGG neural network was trained on a Diagrammatic Image Dataset (DID) to model relationships between building parcels and contextual footprints, enabling efficient integration with design workflows. The generative workflow demonstrated that learning-based methods could effectively suggest massing alternatives and contextual boundaries, highlighting their potential as an alternative to rule-based design methods for capturing complex relationships between design elements.

1. Mass Design Variations from Reconstructed Prediction
2. Prediction Result and Similarity to Actual Shapes
3. The Result of Filter-Applied Feature Map to a Sample Image by Each Convolutional Max-Pooling Layer
4. Data-driven Map of Morphological Types by Building Footprints
5. 45,852 Diagrammatic Image Dataset of Building Mass

SENIOR RESEARCH STUDIO IN ARCHITECTURE II

Details https://jinmorhee.net/jinmorhee_2-multiimage/seniroresearchstudioW24.html

Course Info. ARCH 702, School of Architecture Planning and Landscape, University of Calgary, Winter, 2024

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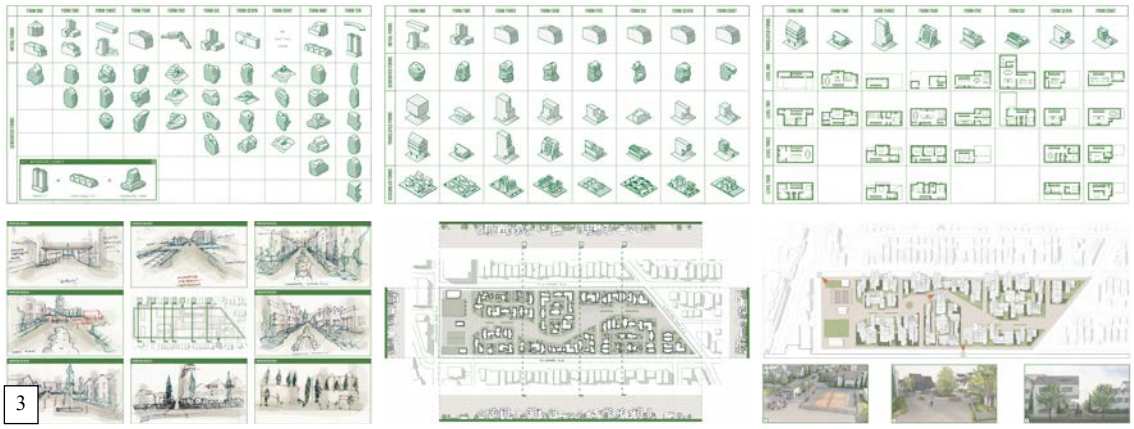


3

This course invites students to delve into the innovative realm of utilizing Artificial Intelligence (AI) in reinterpreting the typology of buildings in Calgary, aiming to foster exploration and the advancement of new design methodologies. This studio stands witness to an era where the explosive growth of AI is increasingly intertwined with architectural and urban design practices. The current framework employed in architectural design involves the incessant generation of images, such as perspectives or plan/section drawings from AI models, transforming these images into 3D forms, and unearthing spatial values from these forms until satisfactory results are achieved.

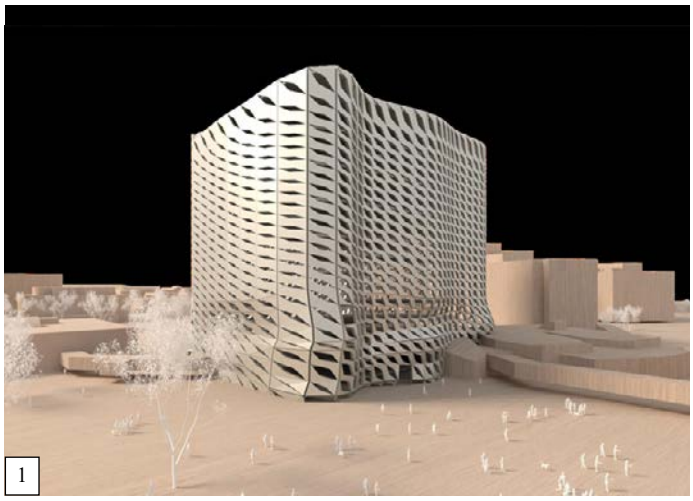
This studio adopts a critical stance on the prevalent tendency to use AI as an oracle in contemporary practices. Instead, it advocates for leveraging AI models creatively and scientifically. The course aims to train AI models on Calgary's building forms to discover new typo-morphologies, reinterpret the city's vernacular, and propose innovative prototypes.

1. Newly Densified Housing Complex with Various Types of Outside Space
2. New High-rise Building Typology for Calgary by Interpolating Typologies of Existing Skyscraper and Multi-public Building
3. A New Type of Vernacular Architecture that Addresses Density, Connection, Community, Privacy, Ownership & Distinction

SCRIPTING AND GENERATIVE DESIGN

Details
https://jinmorhee.net/jinmorhee_2-multiimage/scriptingandgendesF23.html
https://jinmorhee.net/jinmorhee_2-multiimage/scriptingandgendesF23.html
https://jinmorhee.net/jinmorhee_2-multiimage/scriptingparamdesignF22.html
https://jinmorhee.net/jinmorhee_2-multiimage/scriptingparamdesignF21.html
https://jinmorhee.net/jinmorhee_2-multiimage/scriptingparamdesignF20.html

Course Info.
ARCH 702, School of Architecture Planning and Landscape, University of Calgary, Fall, 2023
48-724, School of Architecture, Carnegie Mellon University, Fall, 2020, 2021, 2022, 2023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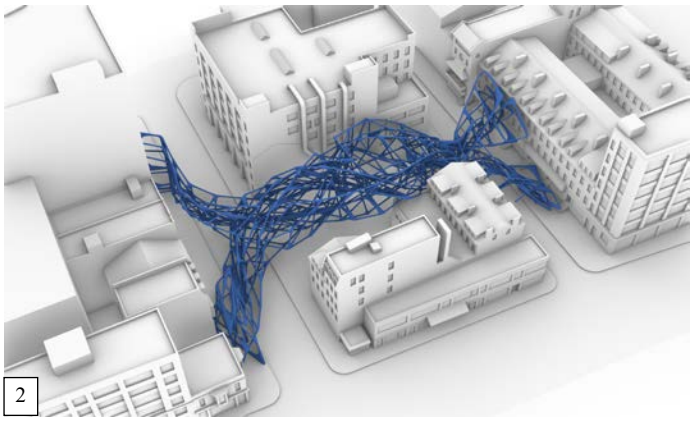
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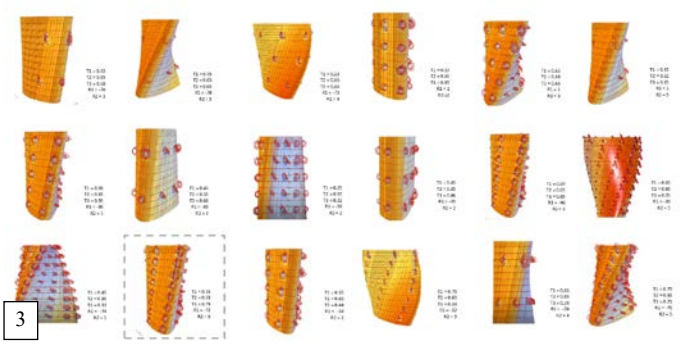
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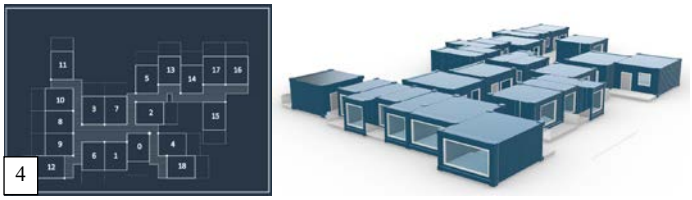
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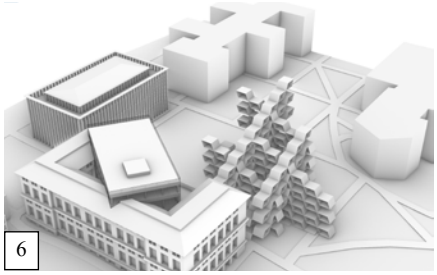
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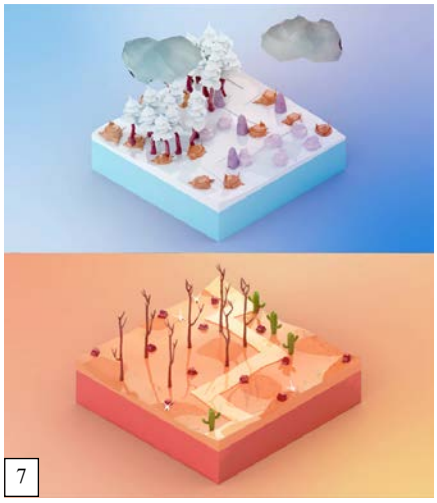
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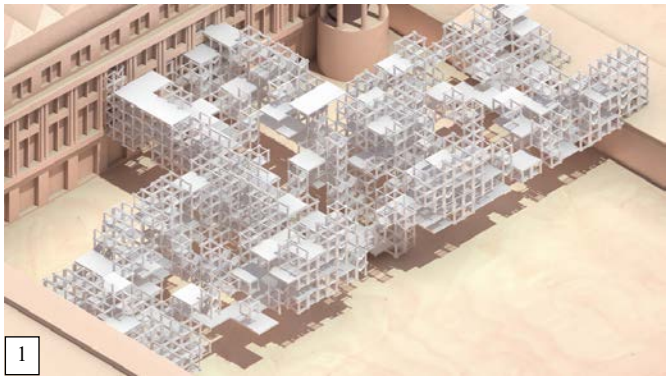
This course prepares students to model geometry for architecture and design applications by scripting data scientific schemes, focusing on form-making algorithms and foundational concepts of generative modeling. Modern architectural geometry modeling is computational, allowing designers greater control over the generative process through variable design constraints, thereby enhancing the exploration of design variations and the analysis of design outcomes. These processes are encapsulated in generative systems, which encompass algorithms and models from fields such as parametric modeling, agent-based modeling, and machine learning. The course aims to equip students with computational skills to formulate design problems and develop generative systems, divided into two parts: the first introduces Python scripting and geometric constructions, while the second applies these techniques through a final project.

1. High-performance Facade Design
2. Final Project, Agent-based Generative Structure
3. Evolutionary Form Finding Using Solar Radiation
4. Optimized Layout Planning for Modular Shelter
5. Bridge Design Using Beaver Nest Algorithm
6. Oblique, Michael Hasey
7. Generative Environment Scene for Animations
8. Recursive Mesh Sculpting
9. Predict Heat Radiation of Floor Plans

GENERATIVE SYSTEMS FOR DEISIGN

Details https://jinmorhee.net/jinmorhee_2-multiimage/gensysdesignS23.html
https://jinmorhee.net/jinmorhee_2-multiimage/gensysdesignS22.html

Course Info. 62-706, School of Architecture, Carnegie Mellon University, Spring, 2022, 2023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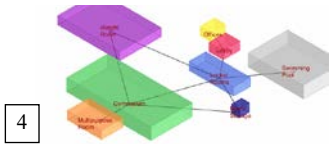
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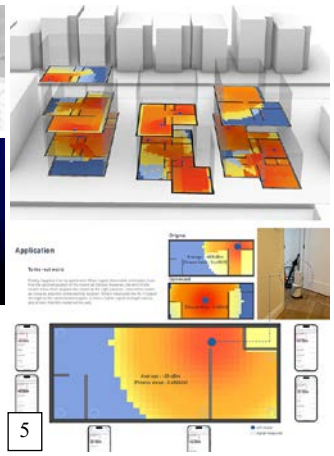
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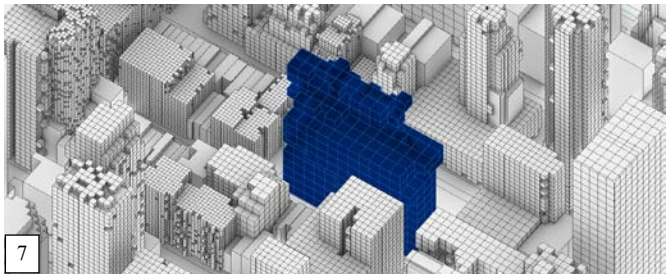
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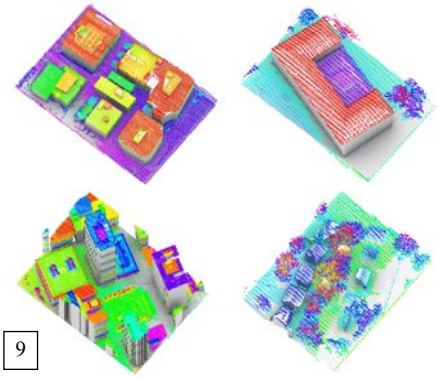
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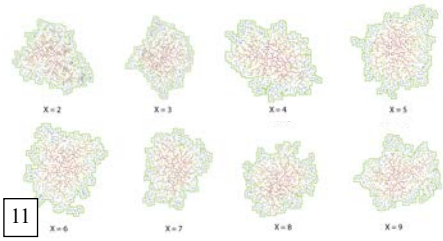
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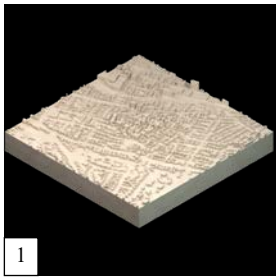
This course is designed for visual artists, enthusiasts, and designers across various fields, such as game design, product design, architecture, and building performance, who wish to explore generative systems—algorithms and computational models that generate design alternatives based on custom inputs. Generative systems, a significant research topic in computational design, have evolved from classical AI and optimization methods to include techniques like parametric modeling, agent-based modeling, and neural networks. This course provides an overview of foundational and contemporary generative techniques, emphasizing the computational formulation of design problems and the synthesis of design alternatives. Students will engage with topics including variational modeling, rule-based modeling, agent-based modeling, and optimization, developing the skills to apply appropriate data structures, algorithms, and models in their projects.

1. Floating Bridge Design
2. The Wind Shaping Tool: Passive Automatic Design Generation Tool
3. The Natural Growth System
4. Conceptual Floor Plans though Visualized Converging Data Graphs
5. Wi-Finder
6. Maze City Generation
7. Voxelized Building Form Generator for Deep Learning
8. Comparision between Latent Space and Parametric Space in Design
9. Various Test Cases for Segmenting Point Clouds and Reconstructing Building Mass, Mass Builder for Urban Point Clouds
10. Parametric Systems for Roof Structures
11. Variations of Generating Urban Networks
12. [G]-Block Tower

MISCELLANEOUS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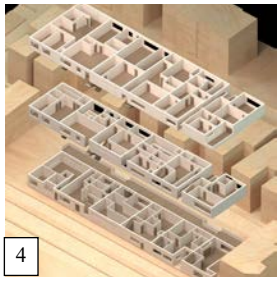
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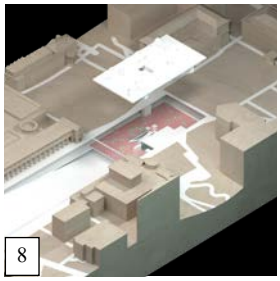
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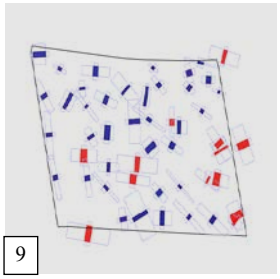
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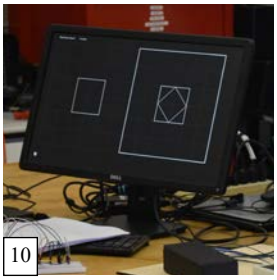
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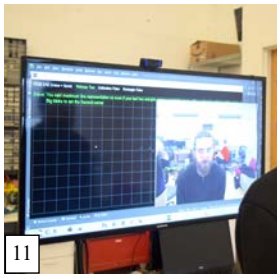
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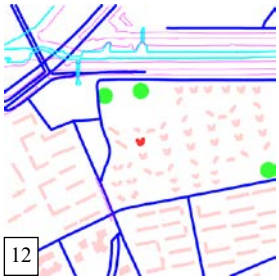
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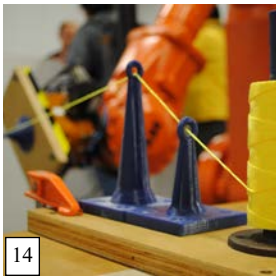
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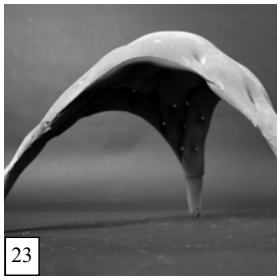
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1. Urban Structure Synthesizer
2. Basic Design Studio, F19
3. Basic Design Studio, F20
4. Introduction into Machine Learning in Design, S21/S22
5. House of Yeesang
6. Youlhyun 201
7. Haedo Elderly WelfaRe Center
8. Underground Plaza Design Using DLA
9. Single Objective Optimization in Apt. Complex Design
10. Reconstruction of Shape Grammar Interpreter
11. VoizeCAD
12. Apartment Price Prediction
13. Massigner
14. Weave
15. Cafe Ch.Gonggam
16. Mapo Social Box Complex
17. Gongreung 1-dong Community Center
18. Gasan Community Center
19. Siheung 5-dong Community Center
20. Ring Bench, Folly Park of Geumcheon Park Renovation
21. Han River Architectural Imagination Exhibition
22. Memories from Ruins, REAL DMZ 2013
23. Computing Aesthetics Of Gravity